

Urban Ogrinec

Young Researcher in Mathematics | Geometric Topology | Applied and Computational Topology

urban.ogrinec@gmail.com | linkedin.com/in/urban-ogrinec | urbanog.github.io

PROFILE

Young researcher in mathematics at the Institute of Mathematics, Physics and Mechanics (IMFM) and doctoral researcher at the University of Ljubljana, Faculty of Mathematics and Physics. My research is in geometric topology, with a focus on topological complexity, LS-category, sectional category and group actions. I enjoy rigorous problem formulation, abstraction, proof, computation and clear technical communication. I am exploring research and development roles where mathematical modeling can support optimization, algorithms, data and AI workflows, and reliable software.

RESEARCH INTERESTS

- Geometric topology of low-dimensional manifolds, knot and link structures, and general position phenomena.
- Topological complexity, LS-category, sectional category, group actions and configuration spaces.
- Applied and computational topology, algorithm design, mathematically grounded data analysis and optimization.

EMPLOYMENT INTERESTS

Interested in research and development work that turns mathematical models into tested software. I would be especially interested in optimization and algorithmic projects, computational experiments, and gradually building stronger skills in Python, backend services, SQL, REST APIs and the team's other tools while contributing to real-world problems.

EXPERIENCE

2023 - present Young Researcher - Institute of Mathematics, Physics and Mechanics (IMFM), Ljubljana

- Research in geometric topology and applied and computational topology; member of the Topology and its Applications group.
- Research themes include topological complexity, sectional category and group actions.

2025 - 2026 Teaching Assistant - Faculty of Mathematics and Physics, University of Ljubljana, Ljubljana

- Teaching support for undergraduate mathematics and communication of abstract mathematical ideas.

2020 - present Personal Assistant - Zavod za samostojno življenje, Ljubljana

- Individual assistance with practical tasks and activities for a person with a severe disability.

2018 Science Researcher - Jožef Stefan Institute, Ljubljana

- Measurements on high-energy particle detectors and organisational support.

EDUCATION

2023 - present Doctoral research in Mathematics - University of Ljubljana, Faculty of Mathematics and Physics, Ljubljana

- Research area: geometric topology and topological complexity.

2020 - 2023 Master's degree in Mathematics - University of Ljubljana, Faculty of Mathematics and Physics, Ljubljana

- Geometry and topology. Thesis: Seiberg–Witten Theory.
- Thesis record: <https://repozitorij.uni-lj.si/IzpisGradiva.php?id=144391&lang=slv>

2020 - 2024 Master's degree in Physics - University of Ljubljana, Faculty of Mathematics and Physics, Ljubljana

- Particle physics. Thesis: Nonlinear Sigma Models.
- Thesis record: <https://repozitorij.uni-lj.si/IzpisGradiva.php?id=159333&lang=slv>

2016 - 2020 Bachelor's degree in Physics - University of Ljubljana, Faculty of Mathematics and Physics, Ljubljana

- Theoretical physics. Thesis: Topological photonics. Final grade: 9.21.

TECHNICAL SKILLS AND LANGUAGES

Programming and mathematical software: Python (intermediate), MATLAB (intermediate), Mathematica (advanced), C++ (beginner), GAP (beginner).

Languages: Slovenian (native), English (C2/C1), German (C1), Italian (B1/B2).

PUBLICATIONS AND PREPRINTS

Parametrized LS category and group actions (2026)

[arXiv:2609.01462](https://arxiv.org/abs/2609.01462) | <https://arxiv.org/abs/2609.01462>

- With Petar Pavešić. Introduces an invariant associated with group actions and develops examples involving mapping tori, pure braid groups and iterated semidirect products.

A Markov Theorem for Braid Closures Up to Link Homotopy (2026)

manuscript submitted September 2026

- A theoretical manuscript on braid moves, link homotopy and algorithmic consequences.

Topological photonics (2020)

Matrika 7(2) | <https://matrika.fmf.uni-lj.si/letnik-7/stevilka-2/ogrinec.html>

- An introduction to Bloch states, Berry phase and curvature, Chern invariants, and applications of topological light.

SELECTED RESEARCH PROJECTS

2022 - 2025 Selected problems in applied and computational topology - ARIS project J1-4001

- Research on topological complexity, persistent fundamental groups and computational approaches to topology.

2022 - 2025 A computational library for knotted structures and applications - ARIS project J1-4031

- Project work connecting knot theory, computational topology, bioinformatics and a Python library for knotted structures.

SEMINARS AND INVITED PRESENTATIONS AT FMF

2024: Parametrized topological complexity, parts 1 and 2; Parametrized topological complexity of sphere bundles, parts 1 and 2.

12 and 19 March 2025: Borsuk-Ulam property and sectional category; Distributional topological complexity and LS-category.

October 2025: Polyhedral products, multi-part series.

12 and 19 November 2025: Sectional category of subgroup inclusions and sequential topological complexity of aspherical spaces as A-genus, parts 1 and 2.

CONFERENCES AND SCHOOLS ATTENDED

7 - 12 November 2024 CIMAT-Mérida Applied Topology School - CIMAT, Mérida, Mexico

9 - 13 February 2026 Homotopy Structures in Barcelona Conference (HoStBCN) - Centre de Recerca Matemàtica, Barcelona, Spain

AWARDS AND ADDITIONAL ACTIVITIES

2019 Dean's Award for Academic Achievement - Faculty of Mathematics and Physics

- Ranked in the top 5% of students, with an average grade above 9.2.

2016 - 2017 Honourable mention - International Olympiad on Astronomy and Astrophysics

- Honourable mention at the 10th and 11th IOAA.

Volunteer and communication: scout leader, choir board member and regular mathematical seminar speaker.